Text Preprocessing Pipeline –25th March

Wednesday, 25 March 2026 6:05 PM

- what is NLP (Natural language processing).?

Sentence (english) → [machine] → generate response

In simple word.

NLP is a field of AI that helps computers to understand, process and generate human language.

- english
- hindi
- Kannada.
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→ Computers do not understand text → they only understand numbers.

* Real world use case
  - Chatbots. (like ChatGPT)
  - Sentiment Analysis.
  - Spam Detection
  - Google search | Auto complete.

I love this movie → [ ] → it see nothing meaningful.

→ How do we convert this text into numbers so machine can understand it

Text → Numbers → machine learning model → understand | predict.

* NLP pipeline. → Structured process.

Raw text → Cleaning → Tokenization → Vectorization → model.

Step-1: Raw text

I love this movie !!! emoji, emoji. → raw text

Step-2 Cleaning the text.

Cleaning → removing noise.

we typically do.

- Lowercasing. I → (i)
- Remove punctuation,

- Remove special characters / emojis

After cleaning

i love this movie. → simple

Step-3 Tokenization

↳ we will break the sentences into smaller units called as tokens.

Eg. i love this move → ['i', 'love', 'this', 'movie'].

why?

Because machine process individual pieces better than a full sentence.

Step-4 Optional stopword removal. sometimes.

an, this, is, the, a

Step-5 Vectorization

↳ converting word into number.

- One Hot encoding.
- Bag of words
- TF-IDF
- Word2vec
- Average2vec

love → 1 0

movie → 0 1

rat cat mat

mat rat cat

run rat cat

dictionary. → set of unique words

	rat	cat	mat	run
rat cat mat	1	1	1	0
mat rat cat	1	1	1	0

Step-6 model.